

Notes: Whited (JF, 1992)

Debt, Liquidity Constraints, and Corporate Investment: Evidence from Panel Data

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1 Big picture

- **Question.** Do imperfections in debt markets affect firms' real investment decisions, especially for firms that are financially weak or informationally opaque?
- **Why this paper matters.** A large earlier literature tested financing constraints by regressing investment on cash flow and q . Whited argues that this reduced-form approach is hard to interpret because average q may be badly measured and because cash flow may proxy for expected future profitability rather than for financing frictions.
- **Core idea.** If a firm faces a borrowing limit, then the marginal value of funds today exceeds the marginal value of funds tomorrow. In a dynamic investment model, that wedge shows up as a **firm-specific shadow cost of external finance**, which raises the effective discount rate relevant for investment.
- **Main theoretical prediction.** The standard neoclassical investment Euler equation should fit **unconstrained** firms, but it should fail for **constrained** firms unless one explicitly adds the effect of financing constraints.
- **Empirical strategy.** The paper estimates an investment Euler equation on a panel of **325 U.S. manufacturing firms** from COMPUSTAT. The usable estimation period is **1975–1986**, though the raw sample spans **1972–1986**. The key empirical move is to augment the Euler equation with a latent financing wedge parameterized by measures of financial distress.
- **Main distress measures.** The two central indicators are **DAR**, the market value of debt relative to the market value of the firm, and **COV**, the interest coverage ratio, defined as interest expense divided by interest expense plus cash flow.
- **Main full-sample result.** The standard Euler equation is strongly rejected in the full sample, but adding the shadow cost of external finance greatly improves fit. In the full sample, the overidentifying-restrictions test falls from $\chi^2 = 34.229$ with $p = 8.14 \times 10^{-5}$ under the standard model to $\chi^2 = 4.091$ with $p = 0.252$ once the financing wedge is included.
- **Key heterogeneity result.** The financing wedge matters more for firms **without bond ratings** than for firms with ratings. The standard model is rejected more sharply for non-rated firms, and the estimated coefficients on the distress variables are larger for them.
- **Key split-sample result.** When firms are sorted by pre-sample financial health, the simple Euler equation fits the healthiest group much better than the most distressed group. For example, after splitting by pre-sample debt-to-assets, the standard equation has $p = 0.400$ for the healthiest group but only $p = 0.012$ for the most distressed group.
- **Main economic interpretation.** Debt-market frictions appear to affect real investment by changing the firm's **effective discount rate**, not just by changing the level of current funds. Financially distressed firms behave as if future investment is heavily discounted relative to current survival and liquidity needs.
- **Main takeaway.** Standard neoclassical investment models miss an important margin: firms differ in access to debt markets, and those differences materially affect the intertemporal allocation of investment.

2 Data and measurement (key objects)

2.1 Sample construction and why it matters

- The paper uses the combined annual and OTC COMPUSTAT industrial files.
- The sample is restricted to **manufacturing firms** with SIC codes between **2000 and 3999**.
- The raw sample period is **1972–1986**, but because the specification uses lags and inventory changes, the estimation sample effectively begins in **1975**.
- Firms with missing or inconsistent data are dropped.
- Firms involved in mergers accounting for more than **15% of assets** are also dropped so that the capital-stock and financing variables are comparable over time.
- The final estimation sample contains **325 firms**: **286** from the combined annual file and **39** from the OTC file.

Interpretation. This is a broad manufacturing panel, but it is not a sample of tiny private firms. If financing constraints show up even here, they may be even stronger in the broader economy.

2.2 Selection and sample-composition caveat

- The retained firms are larger and less levered than the firms dropped from the sample.
- Mean capital stock is about **761.11 million 1982 dollars** for included firms versus **478.17 million** for excluded firms.
- The mean debt-to-assets ratio is **0.339** in-sample versus **0.385** out-of-sample.

Interpretation. The kept sample is somewhat healthier than the discarded one. That likely biases the results *against* finding strong financing constraints.

2.3 Why bond ratings are used as a key split variable

- The paper separates firms by whether they had a **Moody's bond rating** at the start of the sample.
- A bond rating signals prior scrutiny by investors and monitoring institutions, so rated firms should be less affected by informational asymmetries.
- The sample contains **119 firms with bond ratings** and **206 firms without ratings**.
- Rated firms are much larger: the median capital stock is about **441.70 million** versus **26.45 million** for non-rated firms.

Interpretation. A pre-sample bond rating is an institutional proxy for easier access to debt markets and lower asymmetric-information problems.

2.4 Main financial-distress variables

- **Debt-to-assets ratio:**

$$DAR_{it} = \frac{D_{it}}{D_{it} + E_{it}}, \quad (1)$$

where D_{it} is the market value of debt and E_{it} is the market value of equity.

- **Interest coverage ratio:**

$$COV_{it} = \frac{IEX_{it}}{IEX_{it} + CF_{it}}, \quad (2)$$

where IEX_{it} is interest expense and CF_{it} is cash flow.

Interpretation. High DAR means the firm is already heavily levered relative to market value; high COV means debt-service needs are large relative to internally generated funds. Both should be associated with tighter effective borrowing limits.

2.5 Core real-side variables

- **Investment I_t :** spending on plant, property, and equipment, excluding acquisitions.
- **Output Y_t :** sales plus the change in finished-goods inventories.
- **Variable costs C_t :** cost of goods sold plus general, selling, and administrative expenses.
- **Cash flow CF_t :** income plus depreciation.
- **Tobin's q :**

$$Q_t = \frac{D_t + E_t - INV_t}{K_t}, \quad (3)$$

where INV_t is the replacement value of inventories and K_t is the replacement value of capital.

Interpretation. The paper still uses a broad q -style information set through the instrument list, but the estimating equation itself is the Euler equation, not a reduced-form investment-on- q regression.

2.6 Capital measurement and tax-adjusted investment price

- The replacement value of capital is built with a perpetual-inventory method based on gross property, plant, and equipment and investment flows.
- The tax-adjusted price of capital goods incorporates the investment tax credit and the present value of future depreciation deductions.
- The relative price is written as

$$p_t = \frac{(1 - u_t - \tau z_t)P_t^k}{(1 - \tau)P_t}, \quad (4)$$

where u_t is the investment tax credit, z_t is the present value of depreciation deductions, P_t^k is the capital-goods deflator, and P_t is the output deflator.

Interpretation. This is important because the Euler equation compares marginal costs of investing across dates. Taxes and capital-goods prices directly affect those intertemporal costs.

2.7 Descriptive patterns in the summary statistics

- In the bond-rating split, firms without ratings have higher mean leverage and higher mean coverage ratios than firms with ratings.
- In the debt-to-assets split, the most distressed group has mean $DAR = 0.473$, mean $COV = 0.481$, lower mean investment $I/K = 0.164$, and lower mean cash flow $CF/K = 0.250$.
- In the healthiest low-debt group, mean $DAR = 0.211$, mean $COV = 0.171$, mean investment $I/K = 0.212$, and mean cash flow $CF/K = 0.387$.
- Debt growth declines sharply as firms look more distressed, becoming negative for the high-debt group.

Interpretation. The summary statistics line up with the paper's classification scheme: distressed firms are smaller, invest less, generate less cash flow, and expand debt less aggressively.

3 Models and empirical specifications (all equations + interpretation)

3.1 Firm value and capital accumulation without borrowing constraints

The firm chooses investment and borrowing to maximize expected discounted dividends:

$$V_{i0} = E_0 \sum_{t=0}^{\infty} \left(\prod_{j=0}^{t-1} \beta_{ij} \right) d_{it}. \quad (5)$$

The capital stock evolves according to

$$K_{it} = I_{it} + (1 - \delta)K_{i,t-1}. \quad (6)$$

The firm pays dividends out of operating surplus, new borrowing, and investment expenditure, net of taxes and adjustment costs.

Interpretation. In the standard neoclassical model, the firm moves investment across time until the marginal cost of investing today equals the discounted marginal benefit of having more capital tomorrow.

3.2 The unconstrained investment Euler equation

With no binding financing friction, the Euler equation equates current and discounted future marginal investment costs. In words, the firm must be indifferent between:

- investing one more unit today, and
- waiting one period and investing that unit tomorrow.

The unconstrained model implies a standard intertemporal condition of the form

$$\text{marginal cost of investing today} = E_t [M_{t,t+1} \times \text{marginal benefit of waiting until tomorrow}]. \quad (7)$$

Interpretation. If capital markets are frictionless, the firm's effective discount factor is pinned down by market interest rates and taxes, not by its own balance-sheet condition.

3.3 Adding a debt constraint

Whited augments the firm's problem with a borrowing limit:

$$B_{it} \leq B_{it}^*. \quad (8)$$

Let γ_{it} be the multiplier on this debt limit. Then the first-order condition for borrowing becomes

$$(1 + A_{it}) - \beta_{it}(1 + r_{it}^d)E_t(1 + A_{i,t+1}) - \gamma_{it} = 0, \quad (9)$$

where A_{it} is the shadow value associated with the dividend non-negativity condition and r_{it}^d is the tax-adjusted debt return.

Interpretation. A binding borrowing constraint introduces a wedge. The firm now acts as if funds are especially valuable today, so its effective one-period discount rate rises above the market debt rate.

3.4 What financing constraints do to investment

Substituting the borrowing wedge into the investment Euler equation implies that the constrained firm is no longer willing to smooth investment intertemporally in the standard way. Relative to an unconstrained firm, it will behave as if:

- the future is discounted more heavily,
- current liquidity is especially valuable,
- and postponing investment becomes less attractive than the frictionless model predicts.

Interpretation. This is the central mechanism of the paper. Financing constraints matter not because “cash flow enters by hand,” but because balance-sheet weakness changes the intertemporal shadow price of funds.

3.5 Parameterization of marginal product and adjustment costs

To estimate the Euler equation, Whited parameterizes the technology and installation frictions. The marginal product of capital is proxied by output and variable-cost ratios, allowing for non-constant returns to scale and market power:

$$F_K(K_{i,t-1}, N_{it}) = \frac{\eta Y_{it} - \mu C_{it}}{K_{i,t-1}}, \quad (10)$$

where μ is a markup parameter and η is a returns-to-scale parameter.

Adjustment costs are assumed quadratic:

$$\phi(I_{it}, K_{i,t-1}) = \frac{\alpha}{2} \left(\frac{I_{it}}{K_{i,t-1}} - \nu \right)^2 K_{i,t-1}, \quad (11)$$

where α measures the severity of installation costs and ν is a normal investment rate.

Interpretation. The empirical model is still disciplined by theory. The paper is not just throwing financing variables into an ad hoc regression.

3.6 Empirical shadow cost of external finance

Because the latent shadow price is not directly observable, Whited parameterizes it as a quadratic function of distress variables:

$$A_{it} = c_0 + c_1 DAR_{it} + c_2 DAR_{it}^2 + c_3 COV_{it} + c_4 COV_{it}^2. \quad (12)$$

Expected signs are positive: more leverage and weaker coverage should imply a higher shadow cost of external finance.

Interpretation. This is the paper's core measurement move. Instead of using cash flow directly in investment, it lets financial variables move the firm's implied discount rate.

3.7 Estimation method

- The Euler equation is estimated by **GMM**.
- Firm fixed effects are removed by **first differencing**.
- Time dummies absorb aggregate shocks.
- Because differencing creates a moving-average error structure, the paper uses **lagged instruments dated $t - 1$** .
- The instrument set includes lagged investment, output, costs, distress variables, cash flow, taxes, interest expense, depreciation, inventory changes, and Tobin's q .

Interpretation. The overidentifying-restrictions test is the main diagnostic: if the theory is right and the specification is valid, the instruments should be orthogonal to the differenced Euler-equation

error.

3.8 Two core empirical tests

1. **Augmented-versus-standard Euler equation.** Estimate the standard model with $A_{it} = 0$, then estimate the augmented model with A_{it} parameterized by distress variables.
2. **Split-sample validity test.** Sort firms using pre-sample bond ratings or pre-sample distress measures, and ask whether the simple Euler equation fits healthy firms but fails for distressed firms.

Interpretation. This is an unusually sharp design for the early literature. It is not enough that financing variables “matter somewhere”; they should matter *exactly where the theory says they should*.

4 Identification and empirical design (what makes the estimates credible)

4.1 Why the Euler-equation approach is attractive

- It avoids relying on average q as a sufficient statistic for investment opportunities.
- It reduces the classic concern that cash flow is just proxying for omitted future profitability.
- It maps financing frictions into a structural object: the firm’s shadow discount factor.

Interpretation. This is the paper’s biggest conceptual contribution relative to the earlier cash-flow sensitivity literature.

4.2 Why the sample splits are informative

- Firms are sorted using **pre-sample** variables, which helps avoid mechanical endogeneity between the sorting rule and the current Euler-equation disturbance.
- The bond-rating split proxies for external financial access.
- The debt-to-assets and coverage splits proxy for financial distress and debt-market tightness.

Interpretation. The design asks whether the same model works differently across firms that should, on theoretical grounds, face different financing environments.

4.3 What counts as supporting evidence

Support for the financing-constraint interpretation requires three patterns:

- the standard Euler equation should be rejected in the full sample,
- the augmented Euler equation should fit better and assign positive importance to distress variables,
- and the standard Euler equation should fit better for financially healthy firms than for financially distressed firms.

Interpretation. This is a demanding set of predictions. The theory has both a cross-sectional and a specification-test implication.

4.4 Role of fixed effects and time effects

- Firm fixed effects capture time-invariant differences across firms, such as technology, product demand, industry position, and average risk.
- Time dummies capture business-cycle and other aggregate shocks.

Interpretation. This matters because otherwise distress variables might just be standing in for firm type or macro conditions.

4.5 Main limitations

- *DAR* and *COV* are imperfect proxies for the true shadow cost of external finance.
- Bond ratings may also proxy for size and risk, not just information asymmetry.
- The model is built under risk neutrality, so some cross-sectional discount-rate differences could in principle reflect risk rather than financing constraints.
- The sample excludes many smaller and weaker firms, which could matter for external validity.

Interpretation. The paper is strongest on *pattern matching* across groups and specifications, and weaker on pinning down a uniquely causal micro-channel for each distress proxy.

5 Tables and figure

5.1 Table I to Table III: summary statistics and group construction

Read:

- Rated firms are much larger than non-rated firms.
- Non-rated firms have somewhat higher leverage and worse interest coverage.
- In the debt-to-assets split, the high-distress group is clearly smaller, invests less, generates less cash flow, and has negative debt growth.

Table I Summary Statistics: Sample of 325 U.S. Manufacturing Firms, 1972–1986 (Sample Split by the Existence of a Bond Rating)			
Calculations are based on a sample of firms from the COMPUSTAT database. The sample split is based on the pre-sample estimate of a bond rating. Capital stock and market value figures are in millions of 1982 dollars. Investment and cash flow are expressed as a fraction of the capital stock. The interest coverage ratio is defined as the ratio of interest expense to the sum of interest expense and cash flow.			
	Group of Firms		
	Full Sample	Firms with Bond Ratings	Firms without Bond Ratings
Number of firms	325	119	206
Capital stock			
Mean	761.11	1775.45	334.35
Median	31.52	447.70	36.45
Market value of assets			
Mean	1136.21	2085.97	202.95
Median	137.25	1028.66	69.06
Investment			
Mean	0.188	0.183	0.190
Median	0.167	0.167	0.168
Cash flow			
Mean	0.310	0.283	0.324
Median	0.279	0.257	0.294
Sales growth			
Mean	0.074	0.074	0.074
Median	0.084	0.091	0.087
Outstanding debt growth			
Mean	0.032	0.114	−0.011
Median	0.006	0.013	−0.018
Debt to assets ratio			
Mean	0.339	0.330	0.350
Median	0.303	0.282	0.318
Interest coverage ratio			
Mean	0.311	0.258	0.338
Median	0.164	0.157	0.170

Table II Summary Statistics: Sample of 325 U.S. Manufacturing Firms, 1972–1986 (Sample Split by the Debt to Assets Ratio)			
Calculations are based on a sample of firms from the COMPUSTAT database. The sample is split on the basis of the pre-sample level of the debt to assets ratio. Group 1 firms have the lowest debt to assets ratio and Group 3 firms the highest. Capital stock and market value figures are in millions of 1982 dollars. Investment and cash flow are expressed as a fraction of the capital stock. The interest coverage ratio is defined as the ratio of interest expense to the sum of interest expense and cash flow.			
	Group of Firms		
	Group 1	Group 2	Group 3
Number of firms	109	108	108
Capital stock			
Mean	585.19	1144.78	548.42
Median	68.10	79.39	34.34
Market value of assets			
Mean	1596.96	2386.77	612.47
Median	231.42	175.23	79.89
Investment			
Mean	0.212	0.186	0.164
Median	0.187	0.170	0.143
Cash flow			
Mean	0.387	0.294	0.250
Median	0.348	0.266	0.229
Sales growth			
Mean	0.084	0.076	0.052
Median	0.106	0.093	0.082
Outstanding debt growth			
Mean	0.066	0.062	−0.051
Median	0.041	0.036	−0.102
Debt to assets ratio			
Mean	0.211	0.333	0.473
Median	0.171	0.306	0.472
Interest coverage ratio			
Mean	0.171	0.278	0.481
Median	0.115	0.161	0.233

Table III Summary Statistics: Sample of 325 U.S. Manufacturing Firms, 1972–1986 (Sample Split by the Interest Coverage Ratio)			
Calculations are based on a sample of firms from the COMPUSTAT database. The sample is split on the basis of the pre-sample level of the interest coverage ratio. Group 1 firms have the lowest interest coverage ratio and Group 3 firms the highest. Capital stock and market value figures are in millions of 1982 dollars. Investment and cash flow are expressed as a fraction of the capital stock. The interest coverage ratio is defined as the ratio of interest expense to the sum of interest expense and cash flow.			
	Group of Firms		
	Group 1	Group 2	Group 3
Number of firms	109	108	108
Capital stock			
Mean	1035.80	866.01	387.02
Median	31.44	51.84	37.65
Market value of assets			
Mean	1466.79	1242.87	644.22
Median	191.17	175.78	90.44
Investment			
Mean	0.200	0.187	0.175
Median	0.173	0.174	0.152
Cash flow			
Mean	0.371	0.291	0.268
Median	0.333	0.272	0.240
Sales growth			
Mean	0.064	0.078	0.061
Median	0.105	0.086	0.082
Outstanding debt growth			
Mean	0.095	0.039	−0.035
Median	0.068	0.022	−0.014
Debt to assets ratio			
Mean	0.230	0.324	0.461
Median	0.187	0.297	0.437
Interest coverage ratio			
Mean	0.217	0.254	0.489
Median	0.109	0.156	0.244

- In the interest-coverage split, the same broad pattern holds: the most distressed firms are smaller and more fragile.

Economic message. The grouping variables do not look arbitrary. They line up with intuitive differences in financial strength.

5.2 Table IV to Table VI: full-sample and bond-rating-split Euler equations

Table IV GMM Estimates of the Augmented Investment Euler Equation: Full Sample	
The sample consists of 325 U.S. manufacturing firms from the COMPUSTAT database from 1975 to 1986. The Euler equation takes the following form:	
$\left(\frac{1 - \lambda_{it}}{1 + (1 - \tau)j_t - \pi_t^e} \right) \left(\frac{\pi Y_{i,t+1} - \mu C_{i,t+1}}{K_{it}} \right) + \frac{\alpha}{2} \left(\left(\frac{I_{i,t+1}}{K_{it}} \right)^2 - \pi^2 \right) + \alpha(1 - \delta) \frac{I_{i,t+1}}{K_{it}} + (1 - \delta) \frac{p_{i,t+1}}{(1 - \tau)} - \alpha \frac{I_{it}}{K_{i,t-1}} - \frac{p_{it}}{(1 - \tau)} + f_t + s_t = e_{i,t+1},$	

where Y_{it} , C_{it} , I_{it} , K_{it} , and p_{it} are the output, variable costs, investment, capital stock, and price of capital goods of firm i at time t . π_t^e is the expected inflation rate at time t , and i_t is the nominal interest rate paid on corporate bonds at time t . δ is the rate of depreciation for firm i . τ is the corporate tax rate. The shadow cost of external finance, λ_{it} , is parameterized as $c_0 + c_1 DARE_{it} + c_2 DARE_{it}^2 + c_3 COV_{it} + c_4 COV_{it}^2$, where $DARE_{it}$ is the debt to assets ratio and COV_{it} is the interest coverage ratio, which is defined as the ratio of interest expense to the sum of interest expense and cash flow. f_t is a fixed firm effect, which is eliminated by differencing the equation. s_t is a time dummy. Asymptotic standard errors are in parentheses. The standard errors are corrected for the moving average errors induced by differencing using the Newey-West (1987a) procedure.

	$\lambda_{it} = 0$	$\lambda_{it} = c(DARE_{it}, COV_{it})$			
α	2.026**	1.130**	0.975**	1.104**	
μ	(1.010)	(0.490)	(0.245)	(0.329)	
μ	1.344**	0.994**	1.124**	1.108**	
η	(0.214)	(0.112)	(0.257)	(0.136)	
η	0.987	0.842	0.883	0.944**	
π^2	(0.618)	(0.504)	(0.611)	(0.307)	
π^2	0.851	1.449	1.224		
c_1 ($DARE_{it}$)	−	(0.774)	(0.912)	(0.851)	
c_2 ($DARE_{it}^2$)	−	0.022	0.0023	0.0036	
c_3 (COV_{it})	−	(0.0050)	(0.0047)	(0.0091)	
c_4 (COV_{it}^2)	−	0.201**	0.137**	−	
χ^2 (Overidentifying restrictions)	−	(0.094)	(0.188)	−	
Degrees of freedom	−	(0.0022)	(0.011)	−	
p -Value	−	0.028	−	0.054**	
Degrees of freedom	−	0.031	−	0.019	
χ^2 (Exclusion restrictions on λ_{it})	−	0.0081*	−	0.041	
Degrees of freedom	−	(0.040)	(0.033)	−	
p -Value	−	3.70 $\times 10^{-5}$	3.50 $\times 10^{-5}$	5.42 $\times 10^{-5}$	

* Indicates significance at the 5% level.
** Indicates significance at the 1% level.

Table V
GMM Estimates of the Augmented Investment Euler Equation:
Sample of Firms with Bond Ratings

The sample consists of 119 U.S. manufacturing firms from the COMPUSTAT database from 1975 to 1986. The Euler equation takes the following form:

$$\left(\frac{1 - \lambda_{it}}{1 + (1 - \tau)j_t - \pi_t^e} \right) \left(\frac{\pi Y_{i,t+1} - \mu C_{i,t+1}}{K_{it}} \right) + \frac{\alpha}{2} \left(\left(\frac{I_{i,t+1}}{K_{it}} \right)^2 - \pi^2 \right) + \alpha(1 - \delta) \frac{I_{i,t+1}}{K_{it}} + (1 - \delta) \frac{p_{i,t+1}}{(1 - \tau)} - \alpha \frac{I_{it}}{K_{i,t-1}} - \frac{p_{it}}{(1 - \tau)} + f_t + s_t = e_{i,t+1},$$

where Y_{it} , C_{it} , I_{it} , K_{it} , and p_{it} are the output, variable costs, investment, capital stock, and price of capital goods of firm i at time t . π_t^e is the expected inflation rate at time t , and i_t is the nominal interest rate paid on corporate bonds at time t . δ is the rate of depreciation for firm i . τ is the corporate tax rate. The shadow cost of external finance, λ_{it} , is parameterized as $c_0 + c_1 DARE_{it} + c_2 DARE_{it}^2 + c_3 COV_{it} + c_4 COV_{it}^2$, where $DARE_{it}$ is the debt to assets ratio and COV_{it} is the interest coverage ratio, which is defined as the ratio of interest expense to the sum of interest expense and cash flow. f_t is a fixed firm effect, which is eliminated by differencing the equation. s_t is a time dummy. Asymptotic standard errors are in parentheses. The standard errors are corrected for the moving average errors induced by differencing using the Newey-West (1987a) procedure.

	$\lambda_{it} = 0$	$\lambda_{it} = c(DARE_{it}, COV_{it})$			
α	1.734**	1.998**	0.746**	1.238**	
μ	(0.663)	(0.461)	(0.227)	(0.445)	
μ	1.450**	1.109**	1.279**	1.191**	
η	(0.137)	(0.084)	(0.214)	(0.107)	
η	0.598	0.982	1.224	0.669	
π^2	(0.612)	(0.555)	(0.899)	(0.491)	
π^2	1.003	0.882*	0.900		
π^2	(0.440)	(0.513)	(0.871)		
c_1 ($DARE_{it}$)	−	0.0007	0.0005	0.0015	
c_2 ($DARE_{it}^2$)	−	(0.0012)	(0.0021)	(0.0024)	
c_3 (COV_{it})	−	(0.072)	(0.089)	−	
c_4 (COV_{it}^2)	−	0.011*	0.061*	−	
χ^2 (Overidentifying restrictions)	−	(0.007)	(0.038)	−	
Degrees of freedom	−	0.0064	−	0.0084	
p -Value	−	0.0043	−	0.012	
Degrees of freedom	−	0.0074	(0.0050)	−	
χ^2 (COV_{it})	−	0.0064	−	0.0084	
Degrees of freedom	−	0.0043	−	0.012	
p -Value	−	0.0074	(0.0050)	−	
χ^2 (Exclusion restrictions on λ_{it})	21.094	4.577	6.080	10.775	
Degrees of freedom	9	3	5	5	
p -Value	0.013	0.206	0.298	0.065	
Degrees of freedom	16.427	14.854	10.629		
p -Value	−	0.012	0.0049	0.031	

* Indicates significance at the 5% level.
** Indicates significance at the 1% level.

Table VI GMM Estimates of the Augmented Investment Euler Equation: Sample of Firms without Bond Ratings	
The sample consists of 206 U.S. manufacturing firms from the COMPUSTAT database from 1975 to 1986. The Euler equation takes the following form:	
$\left(\frac{1 - \lambda_{it}}{1 + (1 - \tau)j_t - \pi_t^e} \right) \left(\frac{\pi Y_{i,t+1} - \mu C_{i,t+1}}{K_{it}} \right) + \frac{\alpha}{2} \left(\left(\frac{I_{i,t+1}}{K_{it}} \right)^2 - \pi^2 \right) + \alpha(1 - \delta) \frac{I_{i,t+1}}{K_{it}} + (1 - \delta) \frac{p_{i,t+1}}{(1 - \tau)} - \alpha \frac{I_{it}}{K_{i,t-1}} - \frac{p_{it}}{(1 - \tau)} + f_t + s_t = e_{i,t+1},$	

where Y_{it} , C_{it} , I_{it} , K_{it} , and p_{it} are the output, variable costs, investment, capital stock, and price of capital goods of firm i at time t . π_t^e is the expected inflation rate at time t , and i_t is the nominal interest rate paid on corporate bonds at time t . δ is the rate of depreciation for firm i . τ is the corporate tax rate. The shadow cost of external finance, λ_{it} , is parameterized as $c_0 + c_1 DARE_{it} + c_2 DARE_{it}^2 + c_3 COV_{it} + c_4 COV_{it}^2$, where $DARE_{it}$ is the debt to assets ratio and COV_{it} is the interest coverage ratio, which is defined as the ratio of interest expense to the sum of interest expense and cash flow. f_t is a fixed firm effect, which is eliminated by differencing the equation. s_t is a time dummy. Asymptotic standard errors are in parentheses. The standard errors are corrected for the moving average errors induced by differencing using the Newey-West (1987a) procedure.

	$\lambda_{it} = 0$	$\lambda_{it} = c(DARE_{it}, COV_{it})$			
α	0.019**	2.229**	1.126**	1.207**	
μ	(0.694)	(0.674)	(0.332)	(0.813)	
μ	0.014	1.717**	1.082**	1.049**	
η	(0.013)	(0.020)	(0.196)	(0.067)	
η	0.420	0.831	0.988**	0.515	
π^2	(0.507)	(1.335)	(0.161)	(0.319)	
π^2	3.629	1.875	0.846		
π^2	(0.862)	(1.454)	(0.786)		
c_1 ($DARE_{it}$)	−	0.0061	0.0020	0.0072*	
c_2 ($DARE_{it}^2$)	−	(0.0044)	(0.117)	(0.0044)	
c_3 (COV_{it})	−	0.232**	0.442**	−	
c_4 (COV_{it}^2)	−	(0.097)	(0.172)	−	
χ^2 (Overidentifying restrictions)	−	0.0022*	0.013	−	
Degrees of freedom	−	(0.0013)	(0.094)	−	
p -Value	−	0.038**	−	0.055**	
Degrees of freedom	−	(0.012)	−	(0.014)	
χ^2 (Exclusion restrictions on λ_{it})	38.124	0.0188*	0.069	0.069	
Degrees of freedom	9	(0.0079)	6.033	7.394	
p -Value	1.66 $\times 10^{-5}$	0.110	0.193	0.110	
Degrees of freedom	32.091	30.730	29.154		
p -Value	−	6	4	4	
Degrees of freedom	−	1.57 $\times 10^{-5}$	3.48 $\times 10^{-4}$	7.27 $\times 10^{-4}$	

* Indicates significance at the 5% level.
** Indicates significance at the 1% level.

Read:

- **Full sample (Table IV):** the standard model is rejected with $p = 8.14 \times 10^{-5}$. Once A_{it} is parameterized with both distress variables, the model is not rejected, with $p = 0.252$. The exclusion test for the financing variables is strongly significant.
- **Firms with bond ratings (Table V):** the standard model is rejected at $p = 0.013$, but the augmented model is not rejected, with $p = 0.206$. The financing terms matter, but less strongly.
- **Firms without bond ratings (Table VI):** the standard model is rejected much more strongly, with $p = 1.66 \times 10^{-5}$. The augmented model substantially improves fit, with $p = 0.110$. The

coefficients on financial distress are larger.

- A particularly clean comparison is the coefficient on *DAR*: it is about **0.124** for rated firms and **0.252** for non-rated firms in the full augmented specification.

Economic message. Financing frictions are not uniform across firms. They matter much more for firms with weaker access to public debt markets.

5.3 Table VII and Table VIII: does the simple Euler equation fail where it should?

Table VII

GMM Estimates of the Neoclassical Investment Euler Equation: Sample Split by the Debt to Assets Ratio

The sample consists of 325 U.S. manufacturing firms from the COMPUSTAT database from 1975 to 1986. The sample is split into three equally sized groups on the basis of the pre-sample level of the debt to assets ratio. Group 1 firms have the lowest debt to assets ratios and Group 3 firms the highest. The Euler equation takes the following form:

$$\left(\frac{1}{1 + (1 - \tau)i_t - \pi_t^e} \right) \left[\left(\frac{\eta Y_{i,t+1} - \mu C_{i,t+1}}{K_{it}} \right) + \frac{\alpha}{2} \left(\frac{I_{i,t+1}}{K_{it}} \right)^2 \right] + \alpha(1 - \delta) \frac{I_{i,t+1}}{K_{it}} + (1 - \delta) \frac{p_{i,t+1}}{(1 - \tau)} - \alpha \frac{I_{it}}{K_{i,t-1}} - \frac{p_{it}}{(1 - \tau)} + f_i + s_t = e_{i,t+1},$$

where Y_{it} , C_{it} , I_{it} , K_{it} , and p_{it} are the output, variable costs, investment, capital stock, and price of capital goods of firm i at time t . π_t^e is the expected inflation rate at time t , and i_t is the nominal interest rate paid on corporate bonds at time t . δ is the rate of depreciation for firm i . τ is the corporate tax rate. f_i is a fixed firm effect, which is eliminated by differencing the equation. s_t is a time dummy. Asymptotic standard errors are in parentheses. The standard errors are corrected for the moving average errors induced by differencing using the Newey-West (1987a) procedure.

	Group of Firms		
	Group 1	Group 2	Group 3
α	2.046**	0.543**	0.712
(Adjustment cost parameter)	(1.056)	(0.256)	(0.594)
μ	1.075**	0.740**	0.908
(Mark-up)	(0.054)	(0.308)	(0.734)
η	0.894	-0.054	0.071
(Returns-to-scale parameter)	(0.789)	(0.083)	(0.135)
χ^2 (Overidentifying restrictions)	9.417	15.560	21.228
Degrees of freedom	9	9	9
p-Value	0.400	0.077	0.012

* Indicates significance at the 5% level.

** Indicates significance at the 1% level.

Table VIII

GMM Estimates of the Neoclassical Investment Euler Equation: Sample Split by the Interest Coverage Ratio

The sample consists of 325 U.S. manufacturing firms from the COMPUSTAT database from 1975 to 1986. The sample is split into three equally sized groups on the basis of the pre-sample level of the interest coverage ratio, which is defined as the ratio of interest expense to the sum of interest expense and cash flow. Group 1 firms have the lowest interest coverage ratios and Group 3 firms the highest. The Euler equation takes the following form:

$$\left(\frac{1}{1 + (1 - \tau)i_t - \pi_t^e} \right) \left[\left(\frac{\eta Y_{i,t+1} - \mu C_{i,t+1}}{K_{it}} \right) + \frac{\alpha}{2} \left(\frac{I_{i,t+1}}{K_{it}} \right)^2 \right] + \alpha(1 - \delta) \frac{I_{i,t+1}}{K_{it}} + (1 - \delta) \frac{p_{i,t+1}}{(1 - \tau)} - \alpha \frac{I_{it}}{K_{i,t-1}} - \frac{p_{it}}{(1 - \tau)} + f_i + s_t = e_{i,t+1},$$

where Y_{it} , C_{it} , I_{it} , K_{it} , and p_{it} are the output, variable costs, investment, capital stock, and price of capital goods of firm i at time t . π_t^e is the expected inflation rate at time t , and i_t is the nominal interest rate paid on corporate bonds at time t . δ is the rate of depreciation for firm i . τ is the corporate tax rate. f_i is a fixed firm effect, which is eliminated by differencing the equation. s_t is a time dummy. Asymptotic standard errors are in parentheses. The standard errors are corrected for the moving average errors induced by differencing using the Newey-West (1987a) procedure.

	Group of Firms		
	Group 1	Group 2	Group 3
α	1.774**	0.972*	0.078
(Adjustment cost parameter)	(0.686)	(0.534)	(0.056)
μ	1.011**	0.858**	1.024
(Mark-up)	(0.067)	(0.219)	(0.630)
η	0.834	0.0027	-0.017*
(Returns-to-scale parameter)	(0.652)	(0.011)	(0.0079)
χ^2 (Overidentifying restrictions)	8.452	17.771	29.651
Degrees of freedom	9	9	9
p-Value	0.489	0.038	5.03×10^{-4}

* Indicates significance at the 5% level.

** Indicates significance at the 1% level.

Read:

- **Debt-to-assets split (Table VII):** for low-debt firms, the standard Euler equation is not rejected ($p = 0.400$); for the middle group, rejection is borderline ($p = 0.077$); for the high-debt group, the equation is rejected ($p = 0.012$).
- **Coverage-ratio split (Table VIII):** for low-coverage firms, the simple model is not rejected ($p = 0.489$); for the middle group, it is rejected ($p = 0.038$); and for the high-coverage group, rejection is very strong ($p = 5.03 \times 10^{-4}$).

Economic message. This is the sharpest evidence in the paper. The standard Euler equation works best exactly where the theory predicts financing constraints should be least important.

5.4 Table IX and Table X: augmented model for healthy versus unhealthy firms

Read:

Table IX

**GMM Estimates of the Augmented Investment Euler Equation:
Samples of Financially Healthy Firms**

The two samples consist of 109 U.S. manufacturing firms from the COMPUSTAT database from 1975 to 1986. The first includes firms with low debt to assets ratios; and the second consists of firms with low interest coverage ratios, which are defined as the ratios of interest expense to the sum of interest expense and cash flow. The Euler equation takes the following form:

$$\left(\frac{1 - \Lambda_{it}}{1 + (1 - \tau)i_t - \pi_t^e} \right) \left[\left(\frac{\eta Y_{i,t+1} - \mu C_{i,t+1}}{K_{it}} \right) + \frac{\alpha}{2} \left[\left(\frac{I_{i,t+1}}{K_{it}} \right)^2 - v^2 \right] \right] + \alpha(1 - \delta) \frac{I_{i,t+1}}{K_{it}} + (1 - \delta) \frac{p_{i,t+1}}{(1 - \tau)} - \alpha \frac{I_{it}}{K_{i,t-1}} - \frac{p_{it}}{(1 - \tau)} + f_i + s_t = e_{i,t+1},$$

where Y_{it} , C_{it} , I_{it} , K_{it} , and p_{it} are the output, variable costs, investment, capital stock, and price of capital goods of firm i at time t . π_t^e is the expected inflation rate at time t , and i_t is the nominal interest rate paid on corporate bonds at time t . δ is the rate of depreciation for firm i . τ is the corporate tax rate. The shadow cost of external finance, Λ_{it} , is parameterized as $c_0 + c_1 DAR_{it} + c_2 DAR_{it}^2 + c_3 COV_{it} + c_4 COV_{it}^2$, where DAR_{it} is the debt to assets ratio and COV_{it} is the interest coverage ratio. f_i is a fixed firm effect, which is eliminated by differencing the equation. s_t is a time dummy. Asymptotic standard errors are in parentheses. The standard errors are corrected for the moving average errors induced by differencing using the Newey-West (1987a) procedure.

	Low-Debt Firms			Low-Coverage Firms		
α	0.956**	1.051**	1.312**	1.742**	1.244*	1.646**
(Adjustment cost parameter)	(0.222)	(0.302)	(0.632)	0.573	(0.691)	(0.872)
μ	1.033**	1.218**	1.322**	1.567**	1.437**	1.438**
(Mark-up)	(0.078)	(0.149)	(0.181)	(0.439)	(0.694)	(0.407)
η	0.998	0.839	0.814	0.643	0.140	0.632
(Returns-to-scale parameter)	(0.712)	(0.710)	(1.132)	(0.520)	(0.710)	(0.694)
v^2	1.172	1.716	0.870	1.490	0.552	1.005
("Normal" rate of investment)	(0.691)	(1.331)	(0.629)	(1.539)	(0.824)	(0.621)
c_0 (Constant)	0.0009	0.0009	0.0069	0.0099	0.0012	0.0013
	(0.0028)	(0.0016)	(0.0043)	(0.014)	(0.0008)	(0.0218)
c_1 (DAR_{it})	0.100*	0.094	—	0.381	0.102	—
	(0.053)	(0.168)	—	(0.200)	(0.071)	—
c_2 (DAR_{it}^2)	0.0012	0.0073*	—	0.0002	0.0065	—
	(0.0058)	(0.0044)	—	(0.0054)	(0.0046)	—
c_3 (COV_{it})	0.023	—	0.096	0.016*	—	0.055
	(0.038)	—	(0.088)	(0.0087)	—	(0.064)
c_4 (COV_{it}^2)	0.0047	—	0.0092	0.0029	—	0.0014
	(0.0041)	—	(0.013)	(0.0097)	—	(0.0076)
χ^2 (Overidentifying restrictions)	6.959	6.037	8.719	7.503	7.560	8.137
Degrees of freedom	3	5	5	3	5	5
p-Value	0.073	0.303	0.121	0.057	0.182	0.149
χ^2 (Exclusion restrictions on Λ_{it})	2.458	3.380	0.698	0.949	0.892	0.315
Degrees of freedom	6	4	4	6	4	4
p-Value	0.873	0.496	0.952	0.987	0.926	0.989

* Indicates significance at the 5% level.

** Indicates significance at the 1% level.

Table X

**GMM Estimates of the Augmented Investment Euler Equation:
Samples of Financially Unhealthy Firms**

The two samples consist each of 108 U.S. manufacturing firms from the COMPUSTAT database from 1975 to 1986. The first includes firms with high debt to assets ratios; and the second consists of firms with high interest coverage ratios, which are defined as the ratios of interest expense to the sum of interest expense and cash flow. The Euler equation takes the following form:

$$\left(\frac{1 - \Lambda_{it}}{1 + (1 - \tau)i_t - \pi_t^e} \right) \left[\left(\frac{\eta Y_{i,t+1} - \mu C_{i,t+1}}{K_{it}} \right) + \frac{\alpha}{2} \left[\left(\frac{I_{i,t+1}}{K_{it}} \right)^2 - v^2 \right] \right] + \alpha(1 - \delta) \frac{I_{i,t+1}}{K_{it}} + (1 - \delta) \frac{p_{i,t+1}}{(1 - \tau)} - \alpha \frac{I_{it}}{K_{i,t-1}} - \frac{p_{it}}{(1 - \tau)} + f_i + s_t = e_{i,t+1},$$

where Y_{it} , C_{it} , I_{it} , K_{it} , and p_{it} are the output, variable costs, investment, capital stock, and price of capital goods of firm i at time t . π_t^e is the expected inflation rate at time t , and i_t is the nominal interest rate paid on corporate bonds at time t . δ is the rate of depreciation for firm i . τ is the corporate tax rate. The shadow cost of external finance, Λ_{it} , is parameterized as $c_0 + c_1 DAR_{it} + c_2 DAR_{it}^2 + c_3 COV_{it} + c_4 COV_{it}^2$, where DAR_{it} is the debt to assets ratio and COV_{it} is the interest coverage ratio. f_i is a fixed firm effect, which is eliminated by differencing the equation. s_t is a time dummy. Asymptotic standard errors are in parentheses. The standard errors are corrected for the moving average errors induced by differencing using the Newey-West (1987a) procedure.

	High-Debt Firms			High-Coverage Firms		
α	1.056**	1.261**	0.526*	1.123**	1.326**	1.017**
(Adjustment cost parameter)	(0.511)	(0.308)	(0.292)	(0.490)	(0.382)	(0.364)
μ	1.309**	1.108**	0.977**	0.998**	1.113**	1.403**
(Mark-up)	(0.245)	(0.269)	(0.150)	(0.155)	(0.304)	(0.024)
η	1.425*	0.905	0.900	0.805	0.862	0.461
(Returns-to-scale parameter)	(0.794)	(0.533)	(1.112)	(0.522)	(0.889)	(0.411)
v^2	2.764	1.753	0.832	2.622	0.873	0.662
("Normal" rate of investment)	(1.926)	(0.974)	(0.536)	(1.916)	(1.540)	(0.724)
c_0 (Constant)	0.0089	0.0009	0.0031	0.0091*	0.0052	0.0045*
	(0.0062)	(0.0043)	(0.0029)	(0.0051)	(0.0031)	(0.0026)
c_1 (DAR_{it})	0.241**	0.426**	—	0.364**	0.343**	—
	(0.108)	(0.096)	—	(0.147)	(0.167)	—
c_2 (DAR_{it}^2)	0.0025	0.025	—	0.016**	0.103	—
	(0.049)	(0.471)	—	(0.0052)	(0.068)	—
c_3 (COV_{it})	0.094**	—	0.021**	0.055*	—	0.060
	(0.049)	—	(0.0066)	(0.031)	—	(0.054)
c_4 (COV_{it}^2)	0.0004	—	0.081	0.0005	—	0.055
	(0.0003)	—	(0.927)	(0.014)	—	(0.029)
χ^2 (Overidentifying restrictions)	6.958	6.645	9.433	8.997	5.952	8.758
Degrees of freedom	3	5	5	3	5	5
p-Value	0.073	0.248	0.019	0.029	0.311	0.119
χ^2 (Exclusion restrictions on Λ_{it})	14.270	14.583	11.795	20.654	23.699	20.893
Degrees of freedom	6	4	4	6	4	4
p-Value	0.027	0.0056	0.093	0.0021	9.18×10^{-5}	3.33×10^{-4}

* Indicates significance at the 5% level.

** Indicates significance at the 1% level.

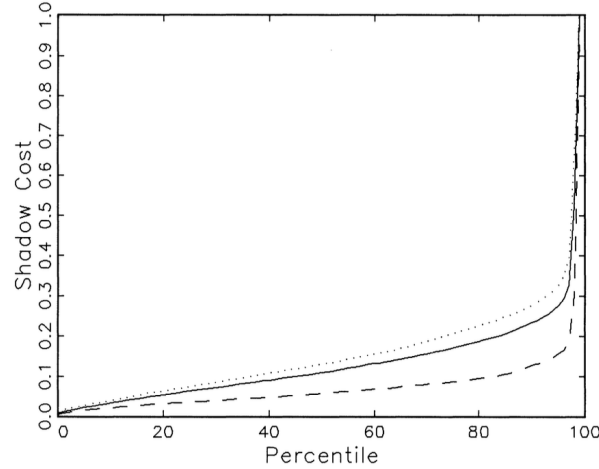


Figure 1. Simulations of the shadow cost of external finance. The sample consists of 325 U.S. manufacturing firms from the COMPUSTAT database from 1975 to 1986. This figure shows the distributions of the simulation of the difference between individual firms' discount rates and the real Baa bond rate in the full sample, and in each of the subsamples divided by the presample existence of a bond rating. The solid line represents the full sample, the dashed line represents the sample of 119 firms with bond ratings, and the dotted line represents the sample of 206 firms without bond ratings. The simulations are calculated using the estimated coefficients from the investment Euler equation, where the shadow cost of outside finance is parameterized as $\lambda_{it} = c_0 + c_1 DAR_{it} + c_2 DAR_{it}^2 + c_3 COV_{it} + c_4 COV_{it}^2$. DAR_{it} represents the firm's debt to assets ratio, and COV_{it} represents its interest coverage ratio, which is defined as the ratio of interest expense to the sum of interest expense and cash flow.

- **Healthy firms (Table IX):** exclusion restrictions on the financial variables generally cannot be rejected. For low-debt firms, the p-values range from **0.496 to 0.952**; for low-coverage firms, from **0.926 to 0.989** in the simpler augmented specifications.
- **Unhealthy firms (Table X):** the financial variables matter a lot. For high-debt firms, the exclusion tests yield p-values such as **0.027** and **0.0056**; for high-coverage firms, **0.0021**, 9.18×10^{-5} , and 3.33×10^{-4} .

Economic message. The financing wedge is not just statistically significant in the pooled sample; it is concentrated where theory says it should be concentrated.

5.5 Figure 1: simulated shadow cost of external finance

Read:

- Using the estimated A_{it} parameters, Whited simulates firm-specific discount-rate premia relative to the real Baa bond rate.
- In each sample, the median premium is below roughly **12%**.
- But the upper tail is very large, approaching **100%** for some firms.
- Firms without bond ratings have a visibly higher distribution of premia than firms with ratings.

Economic message. Financing constraints are not modest uniform wedges. For some firms they are extremely large, which makes the real-investment effects economically meaningful.

6 Short summary

- Whited builds a dynamic investment model in which debt-market frictions raise a firm's effective discount rate.
- She estimates the resulting Euler equation on a panel of 325 manufacturing firms and shows that the standard neoclassical version fails in the pooled sample.
- Once the shadow cost of external finance is parameterized using leverage and coverage ratios, model fit improves dramatically.
- The standard model works much better for healthy firms and much worse for distressed firms.
- The financing wedge is strongest for firms without bond ratings and for firms with high leverage or poor interest coverage.

7 Conclusion

7.1 Core conclusion

The paper provides evidence that asymmetric-information and incentive problems in debt markets affect the real investment decisions of financially weak firms. The standard Euler equation is too restrictive because it assumes a common market discount rate across firms. Once Whited lets that effective discount rate vary with financial distress, the empirical performance of the model improves sharply.

7.2 Economic intuition

A constrained firm cannot freely move resources from the future to the present through borrowing. That makes current internal funds unusually valuable and causes the firm to tilt investment decisions toward current liquidity preservation. The financing friction therefore shows up not as a simple cash-flow term, but as a change in the firm's intertemporal trade-off.

7.3 Takeaway

This paper is an important bridge between neoclassical investment theory and the financing-constraints literature. Its main contribution is to show that debt-market imperfections can be built directly into a structural Euler-equation framework and that doing so explains why financially distressed firms invest differently from healthier firms.